COMSATS University Islamabad, *Lahore Campus*

Department of Computer Engineering

Assignment 1 – FALL 2026

Course Title:	Digital System Design		Course Code:	CPE344	Credit Hours: 4(3,1)
Course Instructor:	Dr. Engr. M Babar Ali		Program Name:	BSc in Computer Engineering	
Semester:	7th	Batch:	FA23	Section:	A/B
Chapters/ Topics:	Ch 1-3	**Due Date:**		**Maximum Marks:**	**30**
Student's Name:		Reg. No.	CUI/ - - /LHR		

Important Instructions / Guidelines:

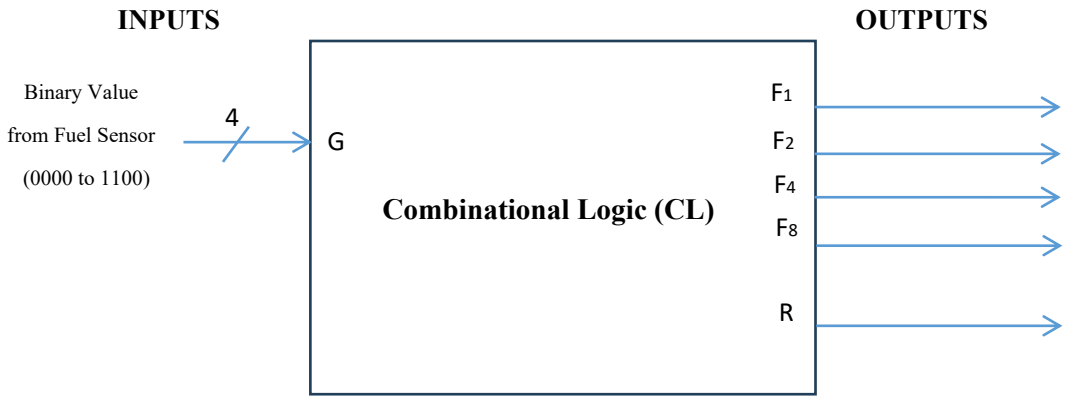
- Write your name and Student registration number clearly on the separate title page (cover page of assignment).
- Answer all the questions.
- Do not give multiple answers to a question.
- Submit handwritten assignment on loose pages. Do not use pulled/ripped out papers from notebook.
- Do not use more than one ink color. Use only blue or black fountain pen or ballpoint pen.
- Do show all intermediate steps while solving a question.
- Plagiarism of work from any source will be penalized and may result in zero out of 30 marks in the assessment.

Question 1: [CLO1, PLO3 – C5] **(Marks 15)**

A vehicle has a fuel sensor which outputs the current fuel/gasoline level as a 4-bit binary number ranges from 0001_2 (0_{10}) to 1100_2 (12_{10}). Design combinational logic (CL) circuit shown in figure below that takes sensor data as input G and generates F1, F2, F4, F8 and R signals on its outputs for fuel indicators accordingly as described below:

- When Gasoline is above 11, CL sets all F_i outputs (where i =1, 2, 4, and 8) to 1 except R which is cleared.
- When Gasoline drops to 11 or below (and stays above 75%), CL sets outputs F_1, F_2 and F_4 to 1.
- When Gasoline drops to 75% or below (and stays above 50%), CL sets outputs F_1 and F_2 to 1
- When Gasoline drops to 50% or below (and stays above 25%), CL sets only output F_1 to 1
- When Gasoline drops to 25% or below, CL clears all outputs to 0
- When the fuel tank level is below 2, CL sets output R only to 1 and clears all F_i outputs (where i =1, 2, 4, and 8) outputs

Implement logic diagram of combinational circuit with five 2 × 4 decoders with enable input.



Question 2: **[CLO1, PLO3 – C5]** **(Marks 15)**

Design a digital system for an automatic door lock with the following I/O signals.

Inputs of the controller:	Outputs of the controller:
• Front gate switch (FS) • Motion detector sensor (MS) • Asynchronous reset switch (R) • Door close switch (DC)	• Front gate melody (FM) • Motion detector melody (MM) • Door open (DO)

The operation of the system is enumerated below:

a. When the reset switch (R) is asserted or the system is first powered on, the FSM goes to the initialization state (***S_init***) immediately.

b. After entering ***S_init***, FSM initializes all output signals to zero and unconditionally moves to the wait state (***S_wait***).

c. From state ***S_wait***, the FSM waits for one of the four switches (***FS, MS, R,*** and ***DC***) to be asserted. All the switches are active high, so when a switch is asserted, it sends out logic 1. The following activities are captured when certain input is asserted:

- **If the button pressed is FS**, the FSM goes to state ***S_front***. In state ***S_front***, the front gate melody is turned on by setting ***FM*** to 1 and the door is opened by setting ***DO*** to 1. The FSM remains in state ***S_front*** until the ***DC*** switch is pressed. Once the ***DC*** switch is pressed, the FSM goes back to S_***wait.***
- **If MS is asserted**, the FSM goes to state ***S_motion***. In state ***S_motion***, the motion detector melody is turned ON by setting ***MM*** to 1 and the door is opened by setting ***DO*** to 1. The ***MM*** and ***DO*** will remain ON for two more clock periods and then it will go back to ***S_wait*** (this state also initializes all outputs).
- **If both FS and MS are asserted**, priority will be given to motion sensor input, so the FSM goes to state ***S_motion***.

d. Pressing the DC switch only affects the FSM when it is in state ***S_front***. The DC switch has no effect on the FSM when it is in any other states.

~~e.~~ From any state, as soon as the reset switch (***R***) is pressed, the FSM immediately (asynchronously) goes back to state ***S_init.***

You are required to create block diagram of the system, capture the FSM and create its architecture.

Good Luck ☺